

Exception handling in java

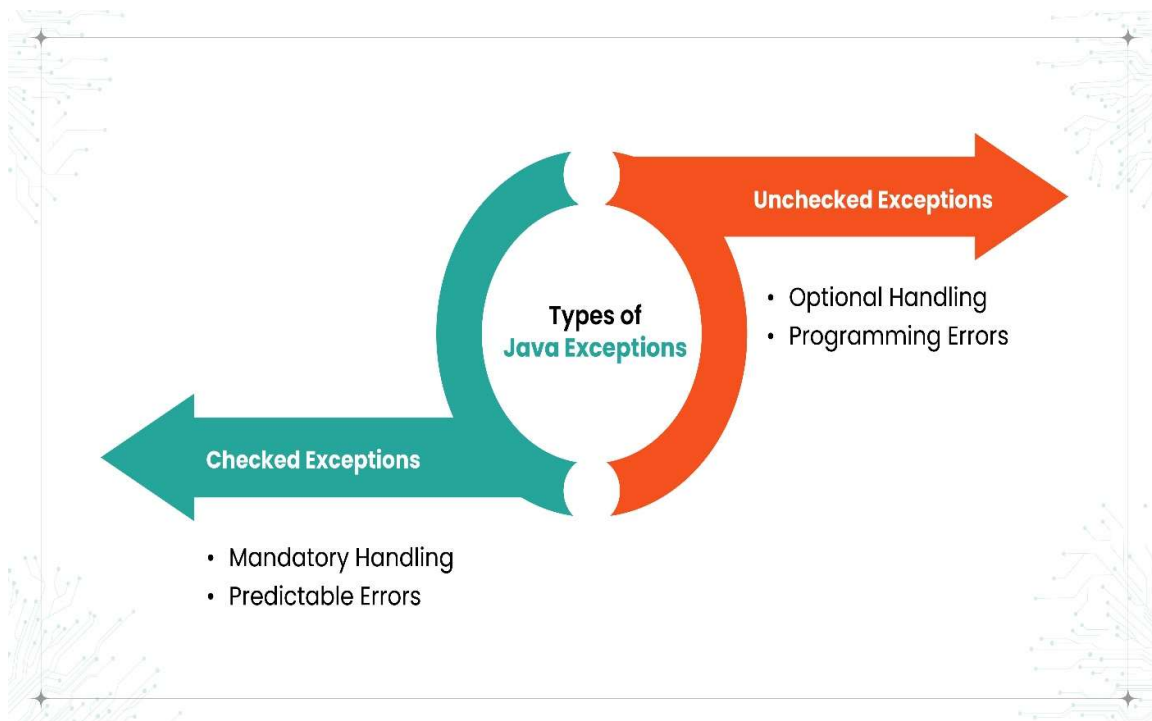
✚ What is exception :-

- An exception in Java is **an error**.
- It is an **unexpected break** in the program's flow that can cause the **program to crash**.

✚ What is exception handling :-

- The Exception Handling in Java is **one of the powerful mechanism** to handle the **runtime errors** so that normal flow of the application can be maintained.

✚ Types of exception handling :-



Checked Exception :-

- Checked Exception are called as a **Compile time exception**.

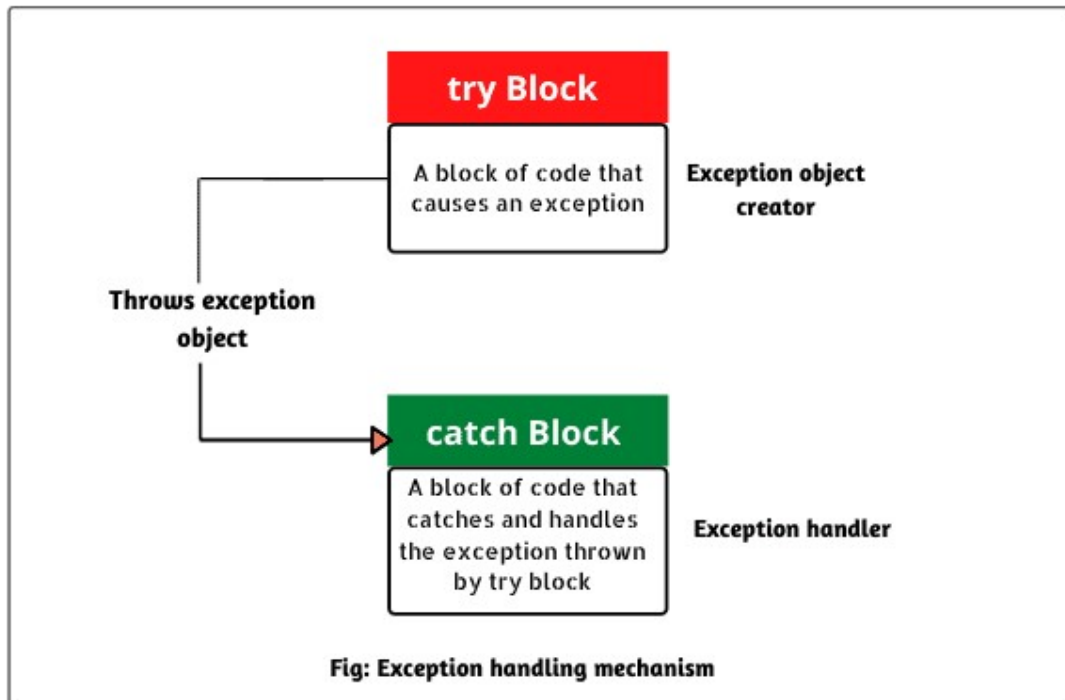
1. **ClassNotFoundException**
2. **InterruptedException**
3. **IOException**
4. **InstantiationException**
5. **SQLException**
6. **FileNotFoundException**

Unchecked Exception :-

- Unchecked Exception are called as a **Run time exception**.

1. **ArithmeticException**
2. **ClassCastException**
3. **NullPointerException**
4. **ArrayIndexOutOfBoundsException**
5. **ArrayStoreException**
6. **IllegalThreadStateException**

✚ Try and Catch Keywords in exception handling :-



✚ Syntax of Try and Catch Keyword :-

```
try
{
    //statement1;
    //statement2;
    //statement3;
    //...
}
catch(exception-type)
{
    //code that handles exception.
}
```

Enclosing statements in try block may throw an exception.

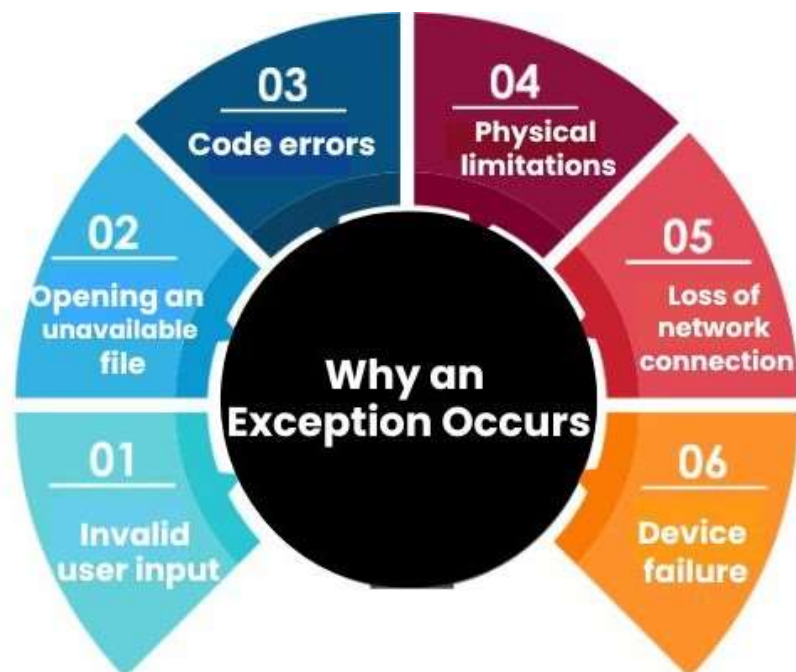
catch block catches and handles a type of an exception.

Try-catch block

+ Importance of Exception Handling :-



+ Why Exception is Occurs :-



Difference Between Exception and Error :-

Aspect	Error	Exception
Definition	Serious problem that can't be fixed by the program.	Problem that can be handled by the program.
Origin	Comes from system failures or hardware issues.	Comes from wrong user input or code mistakes.
Recovery	Cannot be recovered; program stops.	Can be recovered using try-catch.
Types	Syntax, Runtime, Logical errors.	Checked and Unchecked exceptions.
Examples	OutOfMemoryError, StackOverflowError.	IOException, NullPointerException.
Handling	Cannot be handled or caught.	Can be caught and handled.
Hierarchy in Java	Extends Error class.	Extends Exception class.
When They Occur	Can occur anytime (compile or runtime).	Mostly occur at runtime.
Impact	May crash the whole program or system.	Affects flow but can be managed.
Predictability	Unpredictable and out of control.	Can be predicted and handled.

Difference Between Checked and Unchecked Exception :-

Aspect	Checked Exception	Unchecked Exception
Checking Time	Checked during compile time.	Checked during runtime.
Base Class	Derived from Exception.	Derived from RuntimeException.
Cause	Happens due to external issues like file or database errors.	Happens due to coding mistakes or logic errors.
Handling	Must handle using try-catch or throws.	No need to handle compulsorily.
Examples	IOException, SQLException, FileNotFoundException.	NullPointerException, ArrayIndexOutOfBoundsException.

Implementation of Exception Handling

Input :-

```
1 package exception_handling;
2
3 public class handlingarithmeticexception {
4
5     public static void main(String[] args) {
6
7         try
8         {
9             int num=9/0;
10        }
11        catch(Exception e)
12        {
13            System.out.println("Arithmetic Exception" );
14        }
15        System.out.println("Tuesday");
16    }
17 }
```

Output :-

```
Arithmetic Exception
Tuesday
```

Input :-

```
1 package exception_handling;
2
3 public class string {
4
5     public static void main(String[] args) {
6
7         String str = null;
8         try
9         {
10             System.out.println(str.length());
11         }
12         catch(Exception e)
13         {
14             System.out.println(e);
15         }
16     }
17 }
```

Output :-

[java.lang.NullPointerException](#): Cannot invoke "String.length()" because "str" is null

Input :-

```
1 package exception_handling;
2
3 public class character {
4
5     public static void main(String[] args) {
6
7         char ch[]=new char[2];
8         ch[0]='A';
9         ch[1]='*';
10
11         try
12         {
13             ch[2]='O';
14         }
15
16         catch(Exception e)
17         {
18             System.out.println(e);
19         }
20     }
21 }
```

Output :-

[java.lang.ArrayIndexOutOfBoundsException](#): Index 2 out of bounds for length 2